

EJ2201 Electrical Machines and Drives

Examination Assignment Part II

Iraklis Bournazos
Royal Institute of Technology (KTH)
School of Electrical Engineering and Computer Science
`iraklisb@kth.se`

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1 Introduction

In the previous part of examination Assignment, the operation of DC motors was analysed, based on some given lab data. This report focuses on the analysis, the operation and control of two different topologies of AC motors, specifically synchronous and induction machines. It shifts attention to AC machines to explore their characteristics and performance under various conditions.

Key aspects include modeling, parameter identification, and control strategies that optimize energy consumption and improve operational efficiency in these machines.

This report focuses on the analysis, modeling, and control of AC motors, specifically synchronous and induction machines, using laboratory measurements and simulations. The primary aim is to investigate the performance characteristics of these machines and evaluate control strategies that improve energy efficiency and operational stability. The study begins with an exploration of the torque characteristics of synchronous and induction machines by plotting torque as a function of the direct (i_{sd}) and quadrature (i_{sq}) axis currents. These plots, overlapped with the Maximum Torque Per Ampere (MTPA) trajectory, are analyzed to verify consistency with the nameplate data and expected performance.

Key parameters of the three-phase synchronous

and induction machines, obtained from laboratory measurements, are presented alongside the nameplate data. Similarly, the induction machine is modeled using inverse- Γ model. These measured values serve as the foundation for the simulations and control strategies discussed in this report.

Through theoretical modeling, MATLAB simulations, and practical evaluations, the aim is to provide insights into advanced control strategies for AC motors, highlighting the importance of energy-efficient designs and parameter optimization.

2 IPMSM: Torque as a function of i_{sd} - i_{sq} with MTPA Trajectory

In the laboratory, there are seven small 1.5 kW synchronous machines, each characterized by specific nameplate data and laboratory-measured parameters. These data are crucial for simulating and analyzing the machine's performance and for understanding its operational behavior. The following tables summarize the key parameters and the nameplate data of the synchronous machine used in this analysis.

The objective of this analysis is to examine the relationship between the stator currents i_{sd} (d-axis cur-

Parameter	Value
Stator resistance	$R_s = 2.36 \Omega$
d-axis stator inductance	$L_{sd} = 0.0225 \text{ H}$
q-axis stator inductance	$L_{sq} = 0.056 \text{ H}$
Permanent magnet flux linkage	$\psi_{pm} = 0.4990 \text{ Wb}$
Rotor inertia	$J_r = 0.002 \text{ kg} \cdot \text{m}^2$
Viscous friction	$B_r = 1 \times 10^{-3} \text{ N} \cdot \text{m} \cdot \text{s/rad}$

Table 1: Parameters of the three-phase synchronous machine in the laboratory.

Parameter	Value
Rated power	1.5 kW
Rated torque	9.6 Nm
Nominal current (I_N)	3 A RMS
Efficiency (η)	91%
Number of poles	6
Rated speed	1500 rpm
Rated frequency	75 Hz
Back EMF constant (K_E)	1.837 V/s
Torque constant (K_t)	3.18 Nm/A
Back-EMF at 1500 rpm	288 V

Table 2: Nameplate Data of the Synchronous Machine

rent) and i_{sq} (q-axis current) and their influence on the resulting electromagnetic torque. Additionally, the Maximum Torque Per Ampere (MTPA) trajectory is plotted to determine the optimal current distribution that generates maximum torque with minimal current consumption.

To simulate the torque output for different current combinations, a range of values was defined for i_{sd} and i_{sq} . Specifically, the currents were varied from $-1.5 \times i_{max}$ to $1.5 \times i_{max}$, where i_{max} represents the peak current magnitude. This approach allows for the evaluation of torque over a broad range of operating conditions.

The electromagnetic torque was calculated for each point on the current grid using the torque equation for Interior Permanent Magnet Synchronous Motors (IPMSMs), which was provided in lecture 13.

$$T = \frac{3}{2}P(\psi_{pm} \cdot i_{sq} + (L_{sd} - L_{sq}) \cdot i_{sd} \cdot i_{sq}) \quad (1)$$

where:

- P = Number of pole pairs (3)
- ψ_{pm} = Permanent magnet flux linkage
- L_{sd}, L_{sq} = d-axis and q-axis stator inductances
- i_{sd}, i_{sq} = d-axis and q-axis currents

The total current magnitude was calculated as:

$$i_{sdq0} = \sqrt{i_{sd}^2 + i_{sq}^2} \quad (2)$$

The MTPA trajectory minimizes the current magnitude for a given torque. The d-axis and q-axis components corresponding to this trajectory are computed as follows:

$$i_{sd,MTPA} = \frac{\psi_{pm} - \sqrt{\psi_{pm}^2 + 8(L_{sd} - L_{sq})|i_{sdq0}|^2}}{-4(L_{sd} - L_{sq})} \quad (3)$$

$$i_{sq,MTPA} = \sqrt{|i_{sdq0}|^2 + i_{sd,MTPA}^2} \quad (4)$$

To visualize the torque as a function of i_{sd} and i_{sq} , a grid of current values was generated, and the torque equation was evaluated for all possible combinations. This process produced iso-torque curves corresponding to different torque levels of 1 Nm, 2Nm, 3 Nm, 6 Nm, 9 Nm, and the crucial rated torque of 9.6 Nm. The results along with the MTPA trajectory, are presented below:

The calculated maximum torque output of 9.88 Nm corresponds closely to the nameplate value of 9.6 Nm. The value of 9.6 Nm from the nameplate data is achieved when the system reaches the current limit, which is expected as the motor operates at its nominal current. This indicates that the system

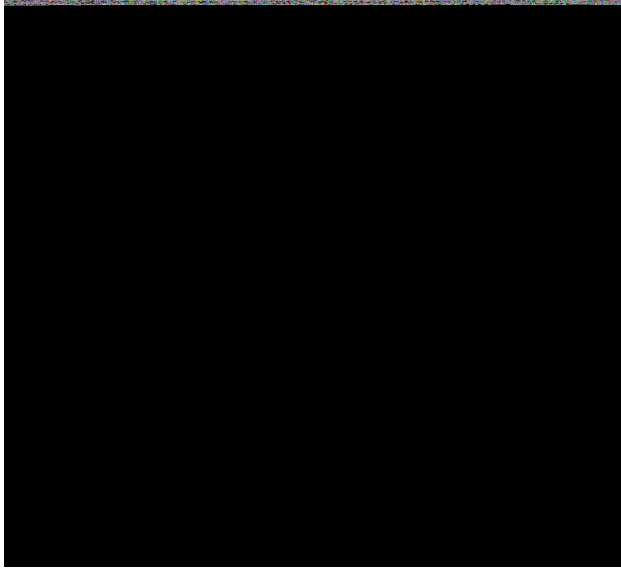


Figure 1: IPMSM-Torque as a Function of i_{sd} and i_{sq} with Overlaid MTPA Trajectory

is working within its designed parameters, with the current limit effectively constraining the maximum torque. Generally, the extracted results are in line with expectations for the motor's performance under typical operating conditions.

3 Induction Machine: Torque as a function of i_{sd} - i_{sq} with MTPA Trajectory

There are seven induction machines in the laboratory, with the nameplate data reported in Table 4 (the first row is the one that refers to the installation). Voltages and currents are in RMS. The inverse- Γ parameters obtained from a converter self-commissioning procedure of one of these machines are reported in Table 3. These parameters are crucial for simulating and analyzing the machine's behavior. The nameplate data of the induction machines, give an overview of their operational characteristics.

In a similar way to the previous question for the

Parameter	Value
Stator resistance	$R_s = 2.97 \Omega$
Leakage inductance	$L_\sigma = 0.029 \text{ H}$
Magnetizing inductance	$L_\phi = 0.343 \text{ H}$
Rotor resistance	$R'_r = 2.66 \Omega$
Rotor inertia	$J_r = 0.0055 \text{ kg} \cdot \text{m}^2$
Viscous friction	$B_r = 1 \times 10^{-3} \text{ N} \cdot \text{m} \cdot \text{s/rad}$

Table 3: Induction machine inverse- Γ model parameters

Table 4: Nameplate of the induction machines in the laboratory

V	Hz	kW	r/min	A	$\cos \phi$	Duty
400 Y	50	1.5	1439	3.2	0.78	S1
230 Δ	50	1.5	1439	5.6	0.78	S1
380 Y	50	1.5	1432	3.3	0.81	S1
220 Δ	50	1.5	1432	5.7	0.81	S1
415 Y	50	1.5	1445	3.2	0.75	S1
460 Y	60	1.5	1748	2.9	0.75	S1

IE3 Efficiency:

IE3-50Hz-85.3%(100%)-84.3%(75%)-81.8%(50%) /
IE3-60Hz-86.5%(100%)

Product code: 3GBP092540-ASK572

Bearings: 6205-2Z/C3, 6204-2Z/C3

Weight: 34 kg

IPMS machine, I chose a specific range for i_{sd} and i_{sq} . More specifically, the currents were defined within a range from -6.78 A to 6.78 A , corresponding to $-1.5 \times I_{i_{\max}}$ to $1.5 \times I_{i_{\max}}$, where $I_{i_{\max}}$ is the peak current magnitude. Specifically, we have:

$$I_{i_{\max}} = 3.2 \times \sqrt{2} \approx 4.52 \text{ A},$$

$$\text{as } I_{\text{peak}} = I_{\text{RMS}} \times \sqrt{2}.$$

Additionally, the torque equation presented in Lecture 15 follows the T-model, but it was necessary to modify it to use the inverse Γ -model.

For induction machines, the equations derived from the equivalent circuit in the inverse Γ -model are as follows:

$$i_r^{g'} = \frac{L_r}{L_m} i_r^g \quad (5)$$

$$\psi_r^{g'} = \frac{L_m}{L_r} \psi_r^g \quad (6)$$

$$L_\sigma = L_s - \frac{L_m^2}{L_r} \quad (7)$$

$$L_\varphi = \frac{L_m^2}{L_r} \quad (8)$$

$$R_r' = \left(\frac{L_m}{L_r} \right)^2 R_r \quad (9)$$

Moreover, based on Lecture 15th the torque equation for the T-model can be expressed as :

$$\tau_r = \frac{3}{2} p \frac{L_m}{L_r} \psi_{rd}^{rf} i_{sq}^{rf} \quad (10)$$

where p denotes the number of pole pairs, L_m is the magnetizing inductance, and L_r is the rotor inductance. The variables ψ_{rd}^{rf} and i_{sq}^{rf} represent the rotor flux linkage and the quadrature-axis rotor current, respectively. The superscript "rf" is used to highlight rotor flux linkage, distinguishing it from fixed stator (s) and rotor position (r).

By substituting the expressions for flux linkage and currents into this formula, we can determine the torque produced in the rotor frame.

However, for the inverse Γ -model representation, the torque equation is simplified and rewritten in terms of the stator reference frame. This approach reduces complexity and highlights the direct (d) and quadrature (q) axis currents. So, By applying

$$\psi_{rd} = L_m i_{sd}$$

and

$$L_\varphi = \frac{L_m^2}{L_r},$$

in the equation of T-model, the torque equation in the inverse Γ -model is:

$$\tau_{\text{inv.}\Gamma\text{-model}} = \frac{3}{2} p L_\varphi i_{sd} i_{sq} \quad (11)$$

Here, L_φ represents the magnetizing inductance, and i_{sd} and i_{sq} denote the direct and quadrature axis

currents, respectively. This formula is particularly useful for visualizing torque production directly from stator currents, simplifying analysis for control design purposes.

Also, by transitioning from the T-model to the inverse Γ -model, the torque expression becomes more intuitive and easier to implement our control theory in machines and it is more practical for applications in field-oriented control systems

To account for Maximum Torque Per Ampere (MTPA) conditions, the torque expression is further adjusted by introducing i_{sdq0} in the modified form:

$$\tau_{\text{inverse } \Gamma\text{-model}} = \frac{3}{2} p L_\varphi i_{sd} \sqrt{|i_{sdq0}|^2 - i_{sd}^2} \quad (12)$$

Next, the derivative of torque with respect to i_{sd} is set to zero:

$$\frac{\partial \tau}{\partial i_{sd}} = 0.$$

The partial derivative is calculated:

$$\frac{\partial \tau}{\partial i_{sd}} = \frac{3}{2} p L_\varphi \left(\sqrt{|i_{sdq0}|^2 - i_{sd}^2} - \frac{i_{sd}^2}{\sqrt{|i_{sdq0}|^2 - i_{sd}^2}} \right).$$

And by solving the equation we have the final MTPA Results:

- $i_{sd, MTPA} = \frac{\sqrt{2}}{2} |i_{sdq0}|$
- $i_{sq, MTPA} = \frac{\sqrt{2}}{2} |i_{sdq0}|$

In a similar way like the synchronous machine we want to generate the Iso-Torque Curves. From the nameplate specifications, the nominal torque τ can be computed as:

$$\tau = \frac{1.5 \times 9550}{1439} = 9.9548 \text{ Nm.}$$

The torque values used for the iso-torque contours are chosen as 1 Nm, 2Nm, 4 Nm, 6 Nm, 8 Nm and the nominal calculated torque value of 9.9548 Nm, which corresponds to the machine's rated operating conditions. Additionally, in the induction machine the calculated machine torque corresponds closely to the nameplate value of the machine. Finally, the extracted results are in line with expectations for the motor's performance under typical operating conditions.

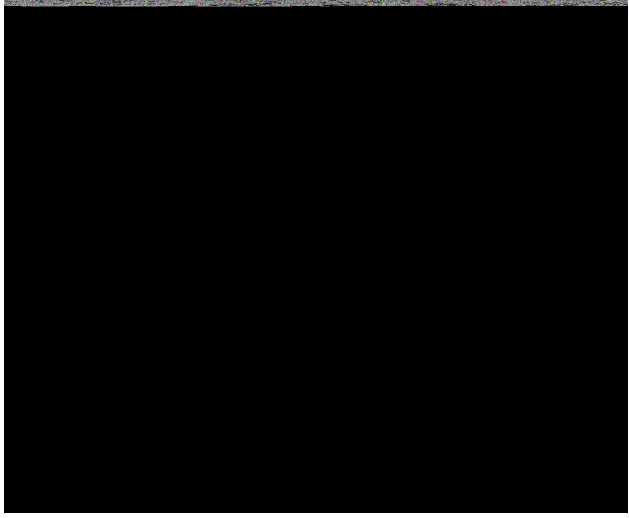


Figure 2: Induction Machine-Torque as a Function of i_{sd} and i_{sq} with Overlaid MTPA Trajectory

4 Non-linearity characteristics on Induction Machines

To accurately capture the *non-linear characteristics* of the induction machine, it is essential to model the *magnetizing inductance* (L_ϕ) as a *variable parameter* instead of treating it as a *fixed constant*. This approach accounts for the effects of magnetic *saturation*, which significantly influences the machine's performance under varying operating conditions.

In this context, the magnetizing inductance (L_ϕ) is redefined to reflect its dependence on the *stator current* (i_{sr}), incorporating both *unsaturated* and *saturated* components. The updated expression for L_ϕ is formulated as:

$$L_\phi(i_s) = \frac{L_{\phi,u} - L_{\phi,sat}}{1 + \left(\frac{i_s}{c}\right)^r} + L_{\phi,sat} \quad (13)$$

where:

- $L_{\phi,u} = 1.5 \cdot L_\phi$ is the *unsaturated magnetizing inductance*.
- $L_{\phi,sat} = L_\phi$ represents the *saturated magnetizing inductance*.

- i_s denotes the *stator current magnitude*.
- $c = 3$ is a *scaling constant* controlling the *rate of saturation*.
- $r = 5$ defines the *order of non-linearity* in the saturation model.

This non-linear formula provides a smooth transition between *linear* and *saturated* magnetic behavior, ensuring that the inductance decreases appropriately as the magnetic flux approaches *saturation levels*.

By modeling the inductance as a *non-linear function*, the machine's dynamic response to changes in current and flux can be captured more accurately. This is particularly important when analyzing scenarios involving *high current demands*, such as *overloads*, *transients*, and *field-weakening operations*.

In the figure below, which was generated using the MATLAB code, we can observe and evaluate the differences between the linear and non-linear approaches. The torque characteristics of the induction machine are analyzed using Iso-torque contours and Maximum Torque Per Ampere (MTPA) trajectories. The Iso-torque curves exhibit some variation. However, the linear and the non-linear MTPA trajectories remain nearly the same.

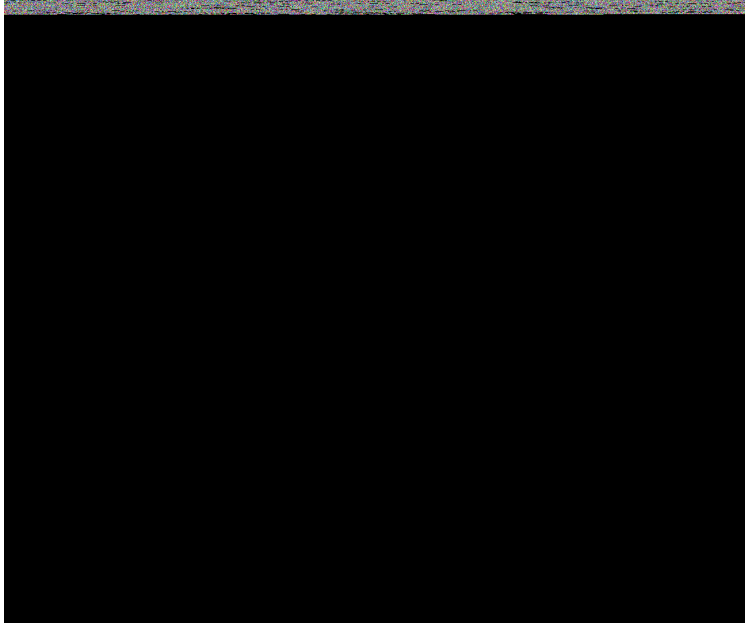


Figure 3: Linear and Non-Linear MTPA Trajectories and Iso-Torque Curves in an Induction Machine

5 Conclusion

We observe that the results provided in the torque analysis for both the synchronous machines and the induction machines are satisfactory. For the simulation, I made efforts and applied some changes. I used the proposed SPMSM model that was provided, I defined the IPMSM parameters with the help of the provided MATLAB code and made some modifications to the model. The adjustments I made included altering the machine block to separate L_s , as for the IPMSM, $L_{sd} < L_{sq}$, and similar changes were necessary in the PI current regulator within the cross-coupling block. For this assignment I used all the information that were provided in the lectures. Finally, I would like to state that discussions with fellow students helped me in some points related to the assignment and I also utilized the assistance of generative AI.